

The Ouroboros Thesis v22 — Convergence

An Honest, Audit-Ready Convergence of the Λ -Aggregator Uniqueness Chain,
Mechanism Truthfulness, and Sim-to-Real Doctrine Transfer

Stephen P. Lutar Jr. · SZL Holdings · ORCID 0009-0001-0110-4173 · 2026-06-03 · CC BY 4.0

Doctrine v11 LOCKED — 749 declarations · 14 unique axioms · 163 sorries @ c7c0ba17 · A5 merged (v11.1 in flight: 794/14/191, axioms unchanged) · Λ -aggregator uniqueness is Conjecture 1 — NOT a theorem · SLSA L1 + L2 attested (NOT L3)

Abstract

v22 ("Convergence") consolidates the May–June 2026 formal-verification advances into the canonical thesis line. Its central correction: the claim that axioms A1–A4 force the weighted geometric mean is false. The asymmetric mean $\Phi(x_1, x_2) = x_1^{2/3} \cdot x_2^{1/3}$ satisfies A1–A4 yet differs from Λ and fails permutation invariance. We add A5 (permutation invariance) as a structure field on LutarAxioms — not a new axiom — keeping the unique-axiom count at 14, and report the partial closure of the n-dimensional Cauchy functional-equation chain that, when complete, would discharge Λ -uniqueness. It is not complete on main; Λ therefore remains Conjecture 1. We additionally report VCG mechanism truthfulness (proven on branch), the achievement of SLSA L2 build provenance (verified on all five GHCR images), the Round 10–11 frontier formalizations, and a Sim-to-Real doctrine-transfer benchmark modeled on the Walrus physical foundation model that measures a mean doctrine α -gap of 0.10 across five unseen compliance regimes. We claim only what is mechanically checked or empirically measured.

1. Recent advances (the substance of v22)

1.1 A5 axiom merge — MERGED (PR #148)

The historical "Theorem 1 — Λ is the unique aggregator under A1–A4" was incorrect. A review of 13 published results (Kolmogorov 1930; Nagumo 1930; Aczél 1948, 1966 Thm 5.1; Hardy–Littlewood–Pólya 1934; Voorneveld 2008) confirms A1–A4 are insufficient. Fix landed on main 2026-06-03: add `IsPermutationInvariant` and an A5 structure field to LutarAxioms; `Lambda_A5_perm_invariant` is sorry-free via `Equiv.prod_comp`. Because A5 is a structure field, the unique-axiom count stays 14; live corpus moves to 794 / 14 / 191.

1.2 Cauchy_ND uniqueness chain — partial closure (IN REVIEW)

With A5 in place, Λ -uniqueness reduces to an n-dimensional Cauchy functional-equation chain: topology / monotone $\rightarrow$ continuity bridge (PR #175, true forms landed), functional analysis (PR #173, closed with 1 honest sorry on the $t=0$ degenerate case), and symmetric branch (PR #174, closed with A5 dependency). The combined chain is not yet complete on main — three PRs open, one residual honest sorry. Therefore Λ stays Conjecture 1.

1.3 VCG mechanism truthfulness — IN REVIEW (PR #172)

`vcgDominantStrategyTruth` and `vcgIndividualRationality` proven on branch via Mathlib's `Finset.exists_max_image` and `add_sum_erase`. Pending merge.

1.4 SLSA L2 achieved — ATTESTED

Build-provenance posture upgraded to L1 + L2; all 5/5 flagship GHCR images (a1loy, sentra, amaru, killinchu, rosie) empirically verified with `slsa-verifier`. SLSA L3 is NOT claimed — hardened/isolated builders

remain roadmap.

1.5 Innovation Rounds 10–11 — IN REVIEW / IN FLIGHT

Round 10: Physics (#177 — Noether, Liouville, Hamiltonian, entropy, A5-from-gauge), Quantum (#176 — PQ signatures, Holevo, Kitaev, ZK, no-cloning→A5), CS (#178 — Byzantine quorum, FLP, CAP), Crypto (#179 — DSSE EUF-CMA, Rekor Merkle, Fulcio, BLS), distsys (PR pending), Round 9 anatomy (#170, 7-organ). Round 11 (formula frontier) in flight. None merged to main yet.

1.6 Sim-to-Real doctrine transfer — Walrus parallel (DESIGN + PARTIAL EMPIRICAL)

Treating the locked doctrine kernel as a pretraining prior and a customer receipt set as fine-tuning data, we define the doctrine α -gap = $|\text{OOD} - \text{in-distribution verdict accuracy}|$. On a live N=60 run against sentra and a11oy, mean α -gap = 0.10: 4/5 unseen regimes transfer at $\alpha=0.00$; adversarial transfers at $\alpha=0.50$ (signature blacklist misses semantically novel attacks). We claim the architecture admits sim-to-real transfer, not that it matches physical-FM accuracy.

Regime	Accuracy	α -gap
R0 control	1.00	—
R1 adversarial	0.50	0.50
R2 cross-jurisdictional	1.00	0.00
R3 multimodal	1.00	0.00
R4 temporal-drift	1.00	0.00
R5 low-data	1.00	0.00
Mean	—	0.10

2. Where v22 sits in the lineage (v1 → v22)

v22 follows v21 ("The PURIQ-OS Substrate", 2026-06-01). It is not a new architecture; it is the convergence of the mathematical-rigor work that v14–v21 deferred. The v19 number was never released (intentional v18 → v20 jump). Concept DOI 10.5281/zenodo.19944926 always resolves to the latest.

Ver	Date	Key contribution
v1	2026-04-28	Ouroboros Loop — looped computation as a system primitive
v3	2026-05-02	The Lutar Invariant Λ — closed-form aggregator
v11	2026-05-11	Applied Λ — measured per-request overhead
v14	2026-05-28	Verifiable multi-agent anatomy; Λ downgraded to Conjecture 1
v18	2026-05-30	Multi-track Substrate Expansion — 29 modules
v20	2026-06-01	The Culmination — formally-verified anatomical substrate
v21	2026-06-01	The PURIQ-OS Substrate — 12-organ runtime, 23 formulas
v22	2026-06-03	Convergence — A5, Cauchy_ND partial, VCG, SLSA L2, R10-11, Sim2Real

Full timeline with per-version DOIs: szl-papers/thesis/THESIS_LINEAGE.md

3. Honest posture

- Λ -aggregator uniqueness is Conjecture 1, conditional on the Cauchy_ND chain closing on main.
- A5 is a structure field, not a new axiom; unique-axiom count remains 14.
- SLSA L1 + L2 attested; L3 not claimed.
- Reed–Solomon $\neq$ holographic. Event-sourcing $\neq$ time travel. Physics analogies are scaffolding.

- Quechua organ names are brand naming, not prior-art or cultural claims. No mystical claims.
- In-review PRs (#170, #172–#179) are labeled as such and not presented as landed theorems.
- Section 889 vendors: Huawei, ZTE, Hytera, Hikvision, Dahua (exactly 5).

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Co-Authored-By: Perplexity Computer Agent <agent@perplexity.ai>